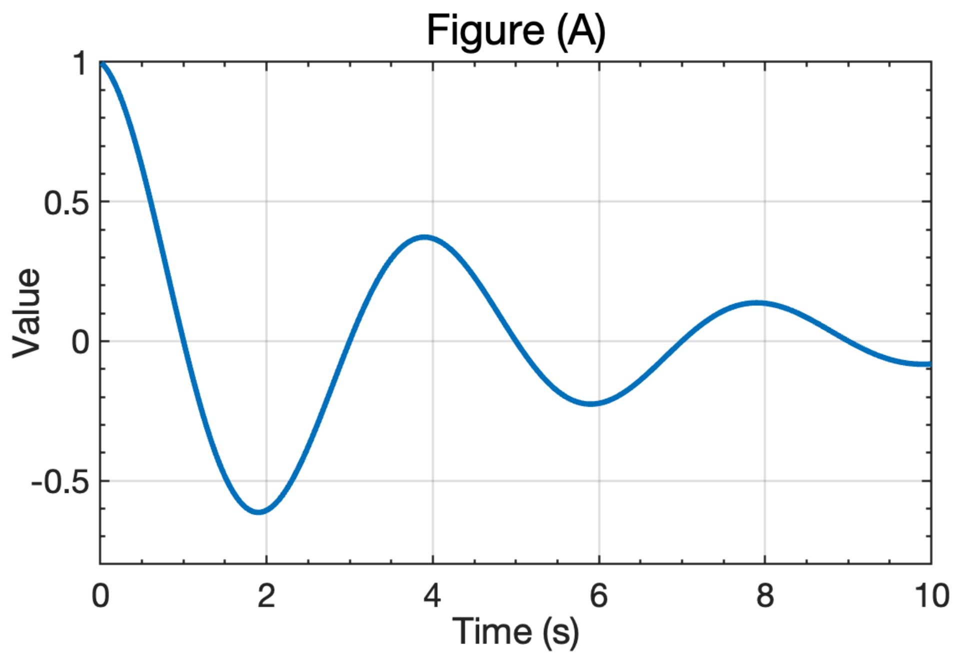
Your grade: 100%

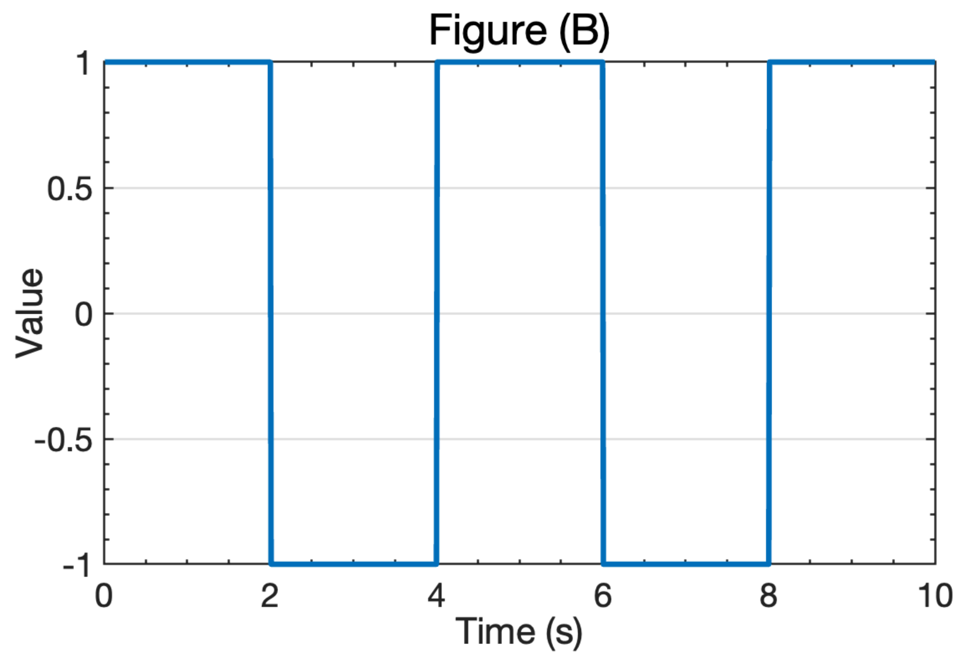
Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

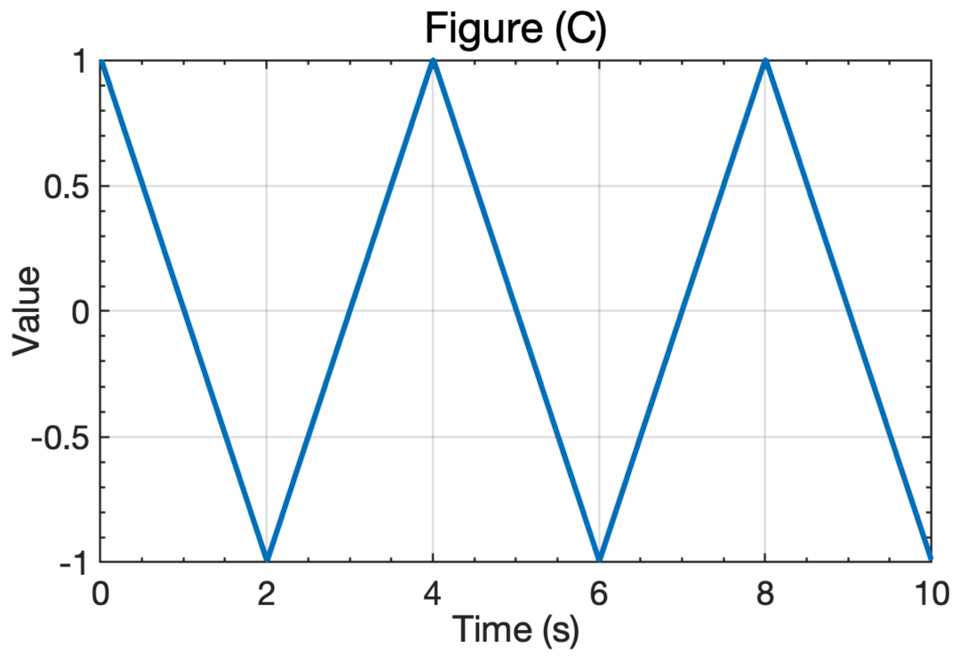
Next item →

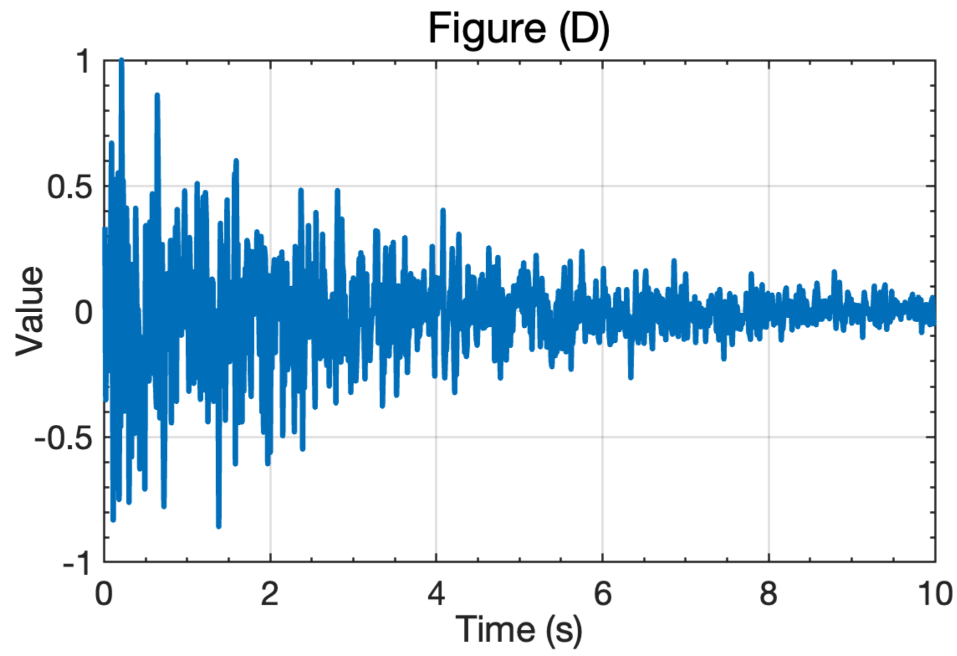
1. Suppose that A is 2×2 in size. Which figure(s) below could be the response e^{At}? (Select all that apply.)

1 / 1 point









- [x] Figure (A).

  Correct
  Yes, this could be the response to e^{At} for some value of A.

- [ ] Figure (B).
- [ ] Figure (C).
- [ ] Figure (D).

2. Suppose that A has eigenvalues -2 and -3. Which of the following is a possible entry of the matrix e^{At}?

1 / 1 point

- ( ) $5e^{+(2+3)t}$.
- ( ) $-2e^{-t} - 3e^{-2t}$.
- (•) $-5e^{-2t} + 5e^{-3t}$.
- ( ) $5e^{-(2+3)t}$.

  Correct
  Yes, this is a possible entry. It has the correct "modes."

3. Suppose that A has eigenvectors:

1 / 1 point

$$v_1 = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 0.3 \\ 0.4 \end{bmatrix}.$$

What transformation matrix T will diagonalize the A matrix via the computation $\Lambda = T^{-1}AT$?

- (•) $T = \begin{bmatrix} 0.1 & 0.3 \\ 0.2 & 0.4 \end{bmatrix}$.
- ( ) $T = \begin{bmatrix} 0.1 & 0.3 \\ 0.2 & 0.4 \end{bmatrix}^{-1}$.
- ( ) $T = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}^{-1}$.
- ( ) $T = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}$.

  Correct
  Yes. This is correct.